

Target and stitched multiple-shooting estimate

actual common support: 19.009–23.959 s

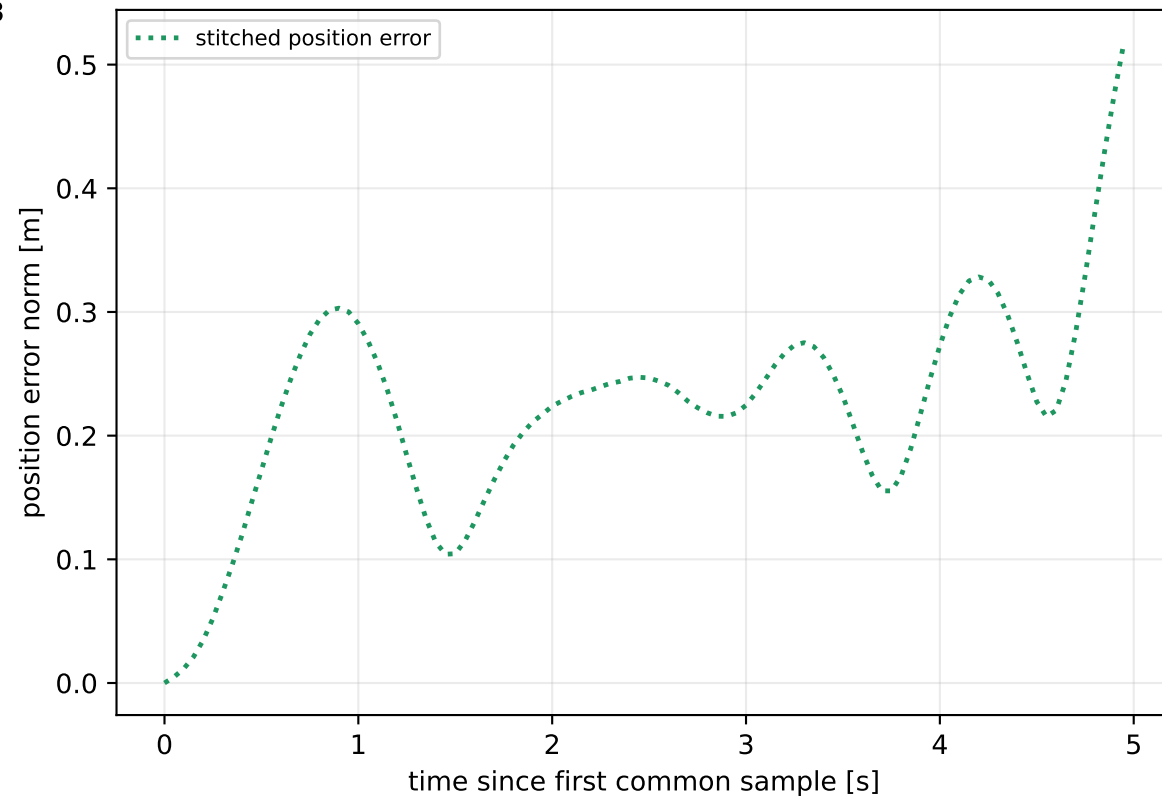
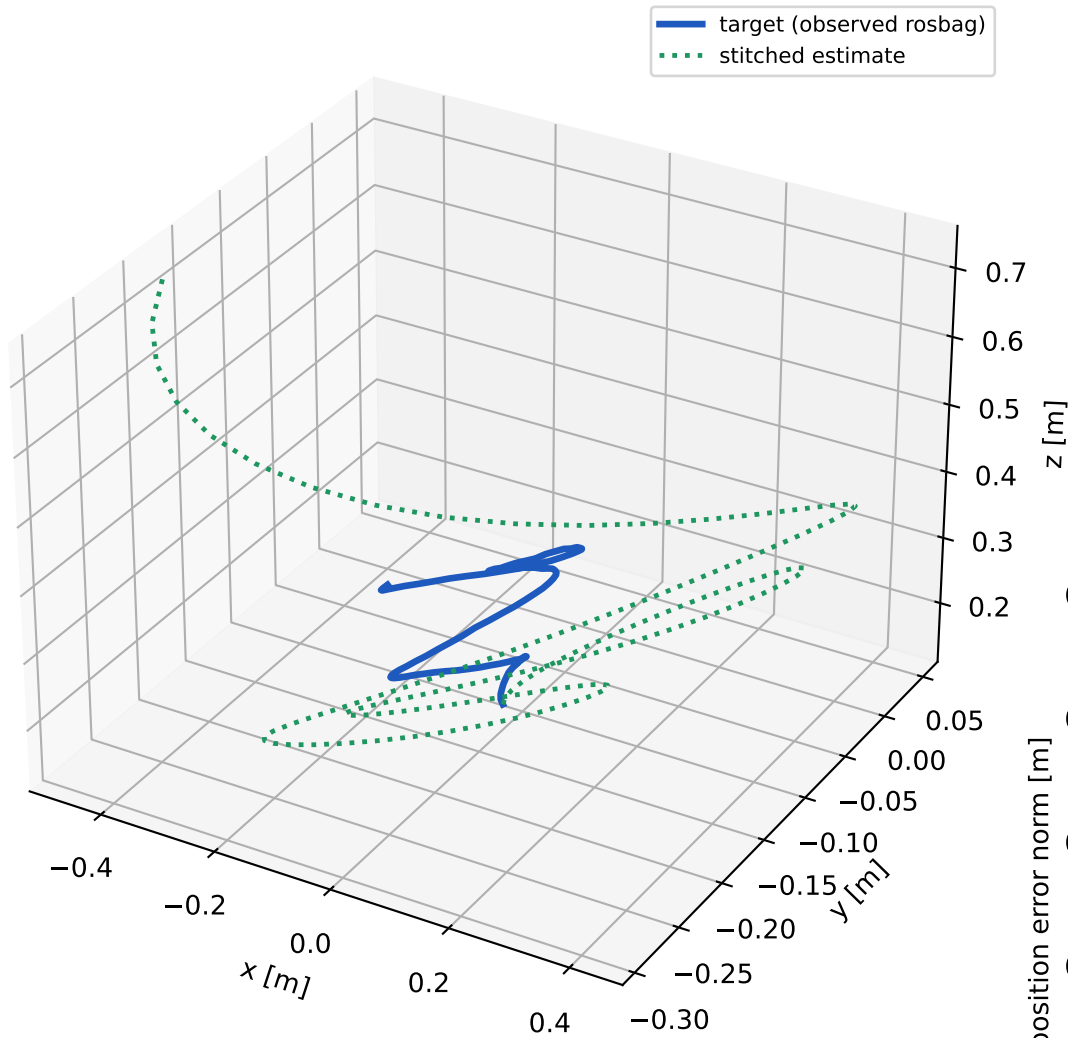
selected command delay: 0.16 s

metric	stitched
position RMSE [m]	0.23911
orientation RMSE [deg]	12.756
velocity RMSE [m/s]	0.67337
angular velocity RMSE [rad/s]	0.76211
specific force RMSE [m/s ²]	1.6015

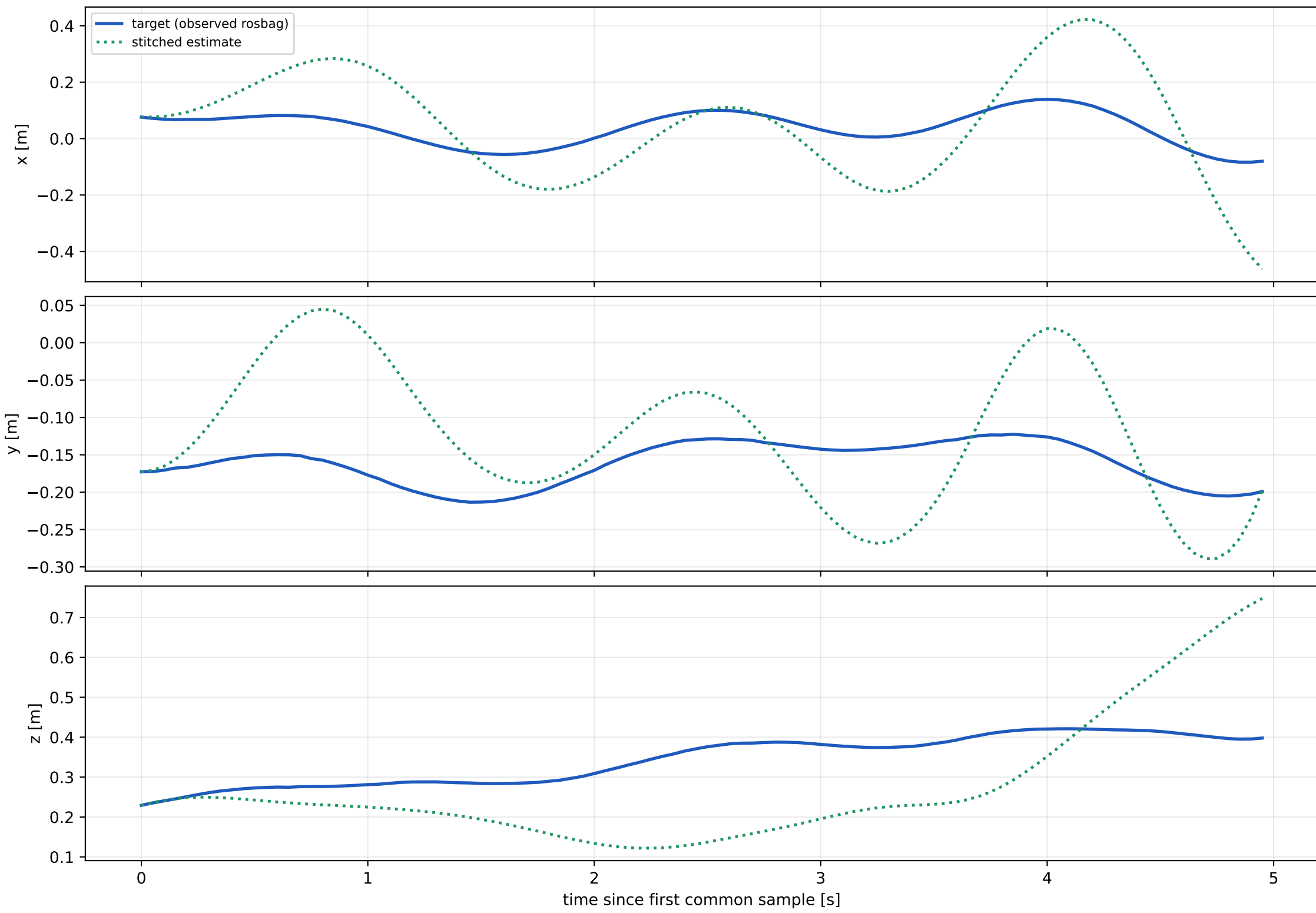
continuity max / tolerance:
0.0002267 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;
the fit uses sensor position and orientation.

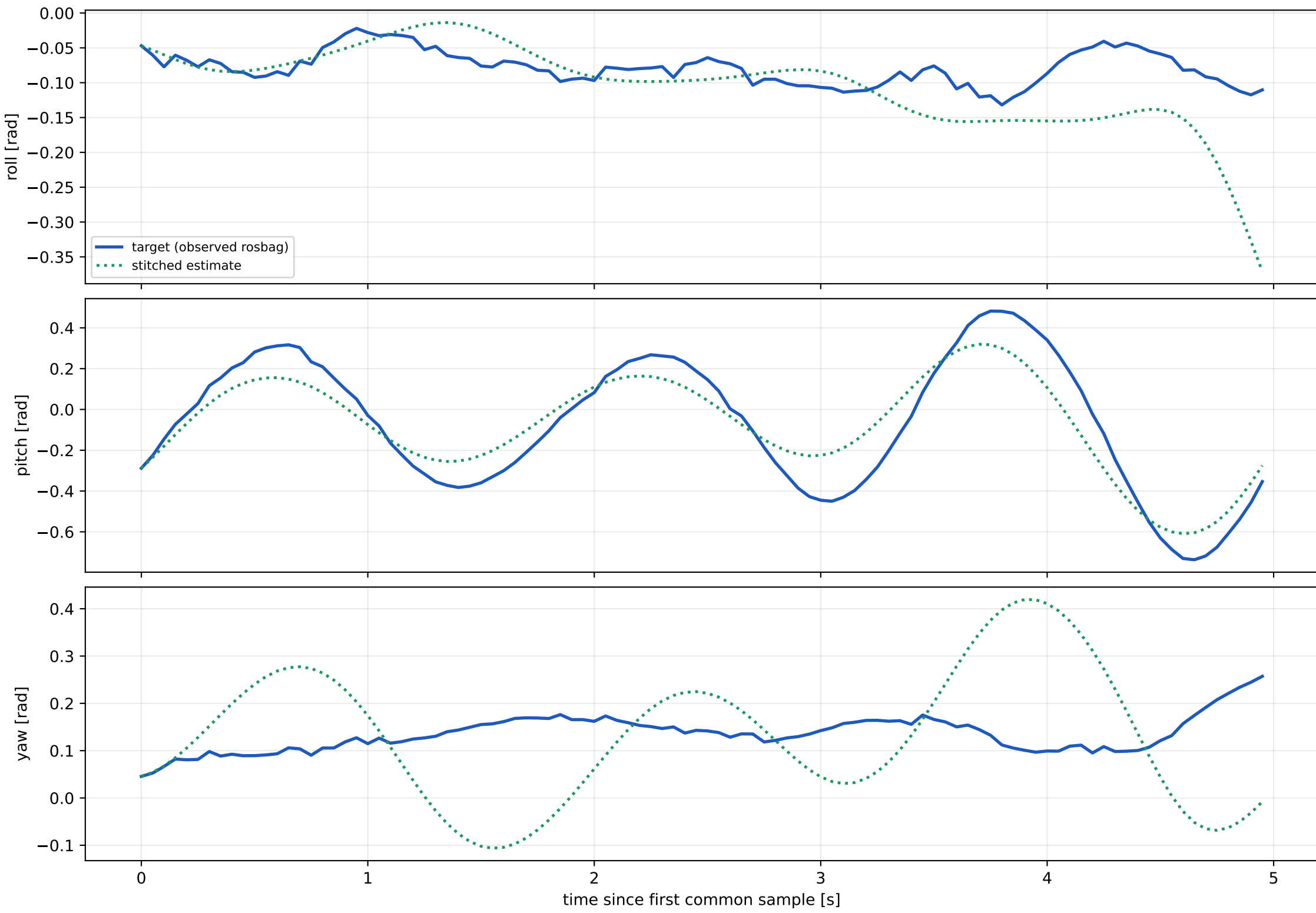
Stitched trajectory



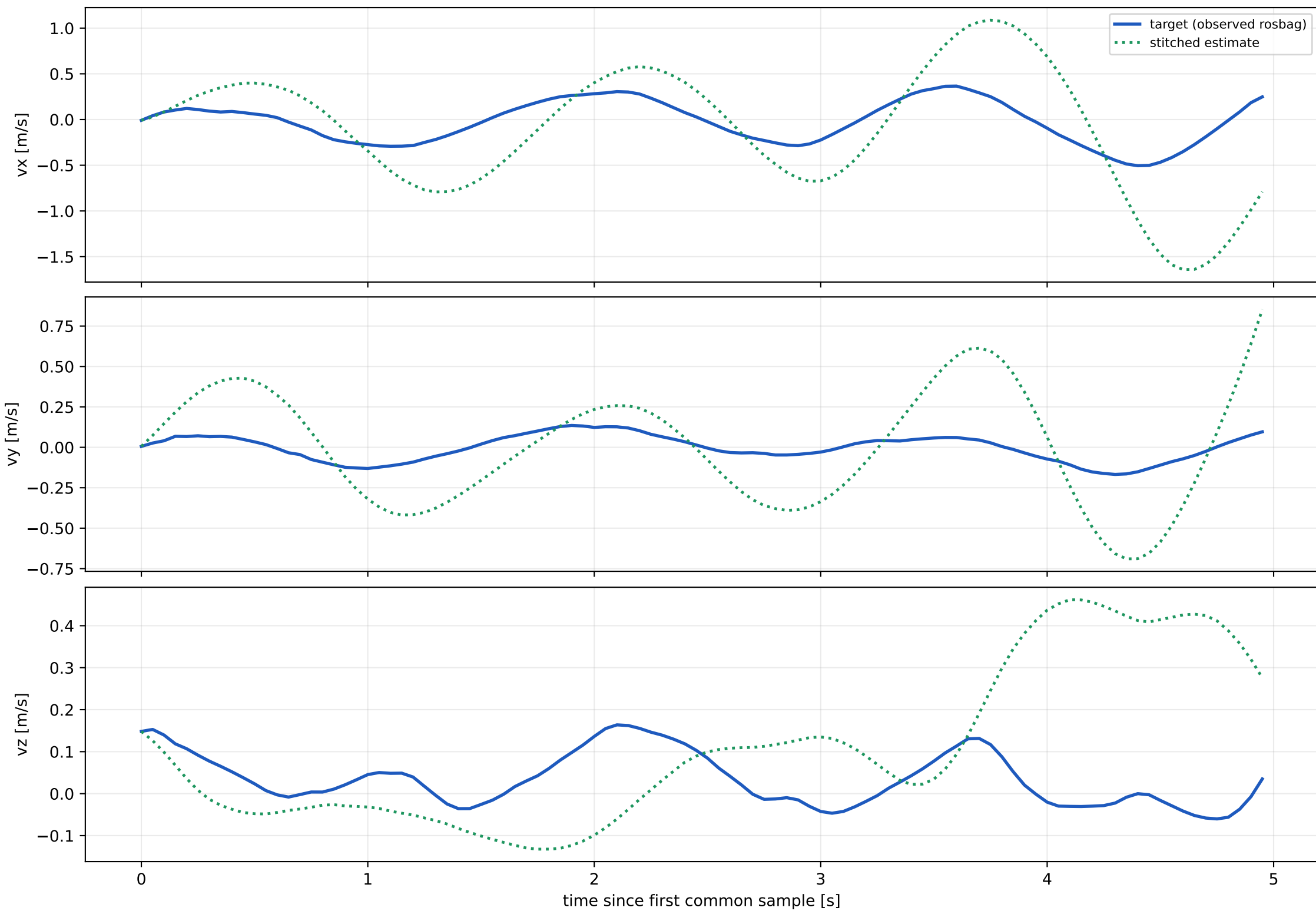
Sensor position



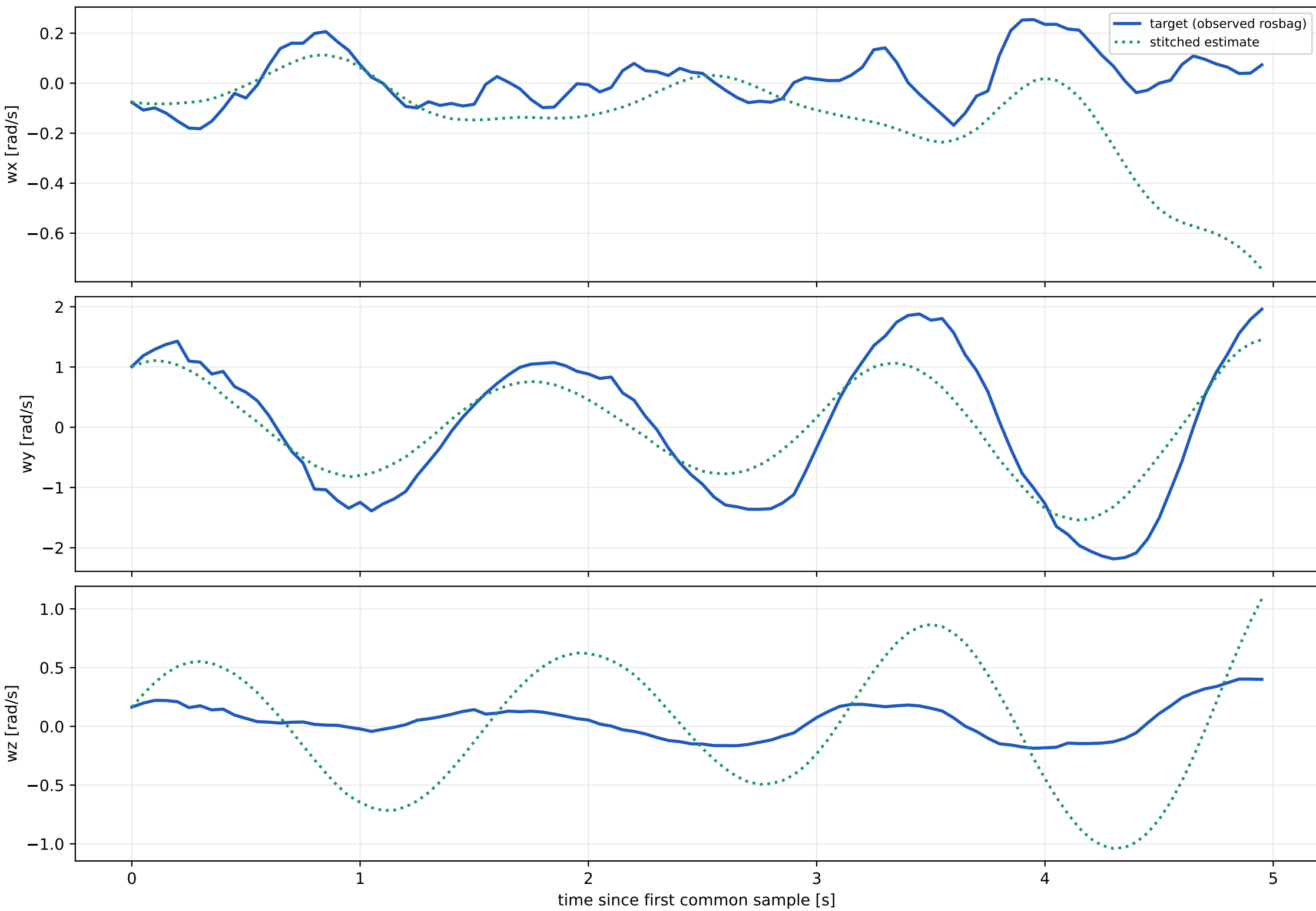
Sensor orientation (display only; fit uses SO(3) log)



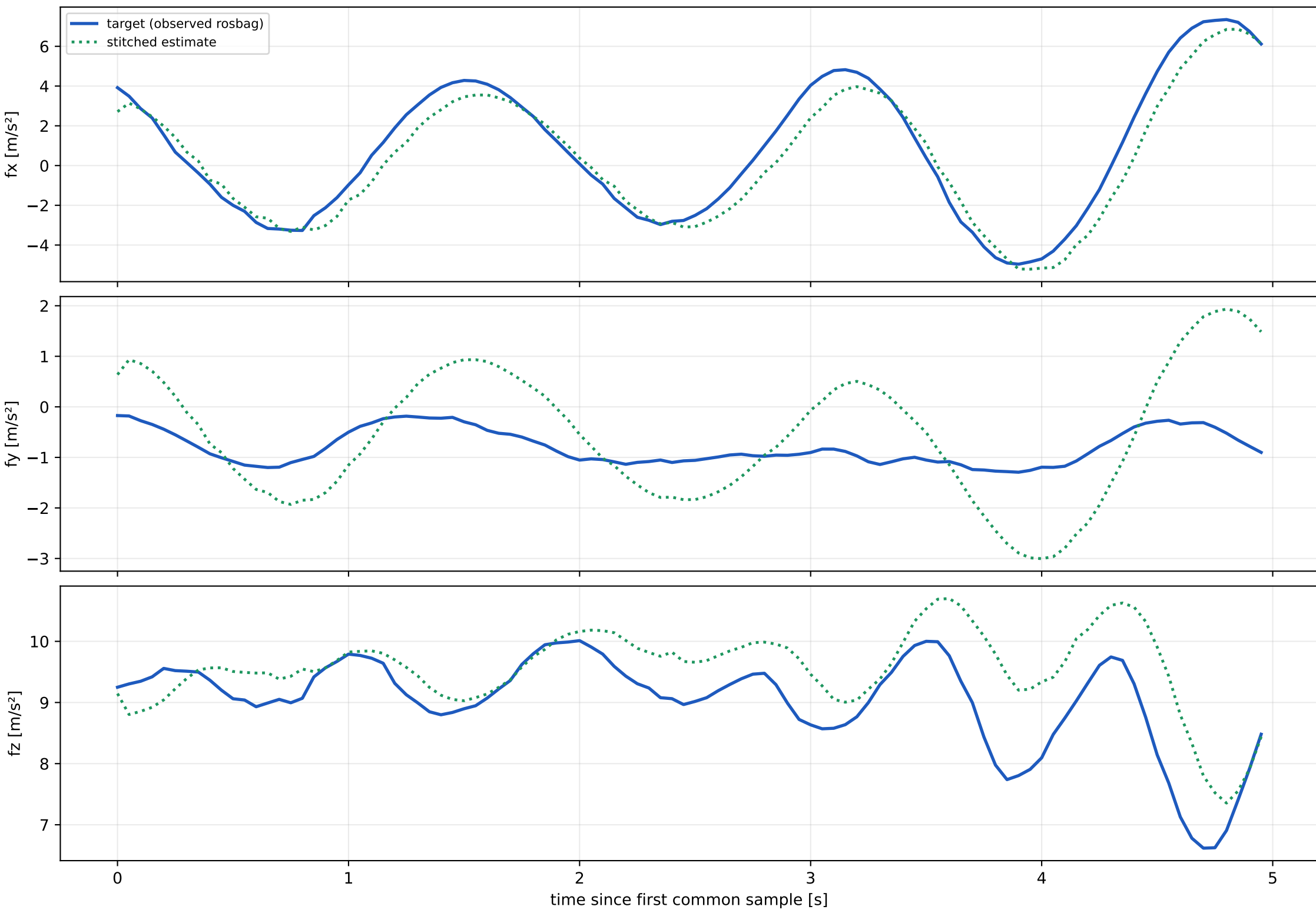
World-frame sensor velocity (diagnostic; excluded from fit)



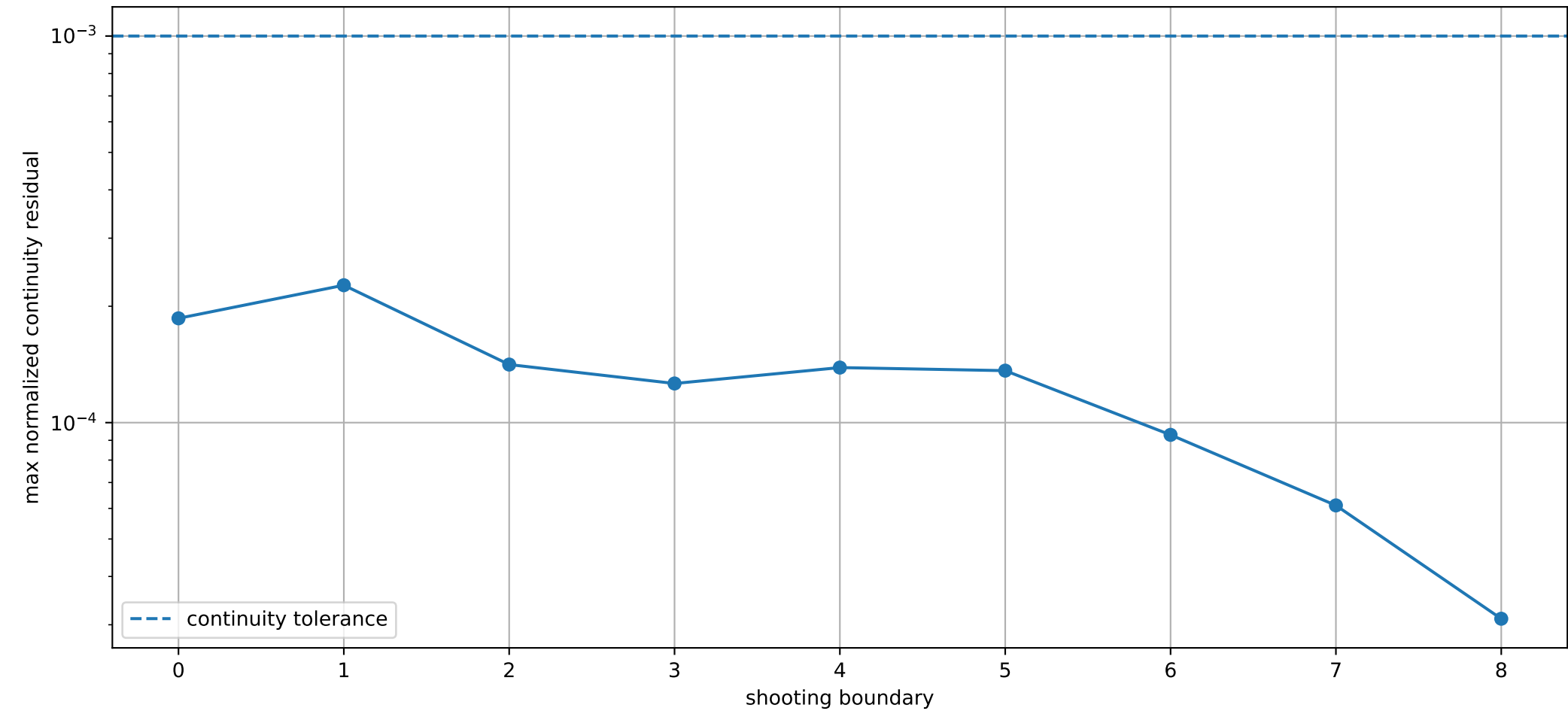
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



Final estimated parameters

Selected multiple-shooting estimate

Decoded physical parameters

```
mass [kg] 3.84104976726
inertia [kg m^2]
[ 0.63761226048  0.141536865451  0.0115772554458]
[ 0.141536865451  0.333363134276 -0.131528139808]
[ 0.0115772554458 -0.131528139808  0.910327812829]
principal inertia [kg m^2] [0.253179324687, 0.687934270593, 0.940189612306]
CoG offset [m] [0.0567211766946, -0.153819661485, 0.0139067227192]
rotor force effectiveness [1.5600382033, 3.50895017732, 0.293416033936, 0.622592266611]
command delay [s] 0.16
```

Thirteen optimized smooth coordinates

```
log_mass_scale 0.490650783072
log_second_moment_cholesky_xx_scale 0.773813075228
log_second_moment_cholesky_yy_scale 1.06339260685
log_second_moment_cholesky_zz_scale 0.00940676677767
normalized_second_moment_cholesky_yx_offset -1.01241447388
normalized_second_moment_cholesky_zx_offset -0.280922042114
normalized_second_moment_cholesky_zy_offset 2.29781388695
cog_offset_x_m 0.0567211766946
cog_offset_y_m -0.153819661485
cog_offset_z_m 0.0139067227192
force_effectiveness_contrast_1 -0.57318541504
force_effectiveness_contrast_2 1.69519172413
force_effectiveness_contrast_3 0.547170371794
```

Fit diagnostics

```
stitched SE(3) loss 13.9600342501
full-rollout SE(3) loss 13.9686854109
broad soft-prior cost 2.23055699989
position RMSE [m] 0.239105735735
orientation RMSE [deg] 12.7556294639
velocity RMSE [m/s] 0.673372647393
angular velocity RMSE 0.76211382569 rad/s
specific force RMSE 1.60150485307 m/s^2
continuity max (normalized) 0.000226678930149
continuity tolerance 0.001
continuity converged True
```